

MODULE 1

EXPONENTIAL FAMILY AND LINEAR MODELS

1.1 PRINCIPLES STATISTICAL MODELING

REGRESSION ANALYSIS · DATA SCIENCE ENGINEERING · ITESO
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Two things that get confused

Science of statistics

Field that includes the collection, analysis, interpretation, presentation and organization of data.

It provides the theoretical foundation and methodologies for working with data.

Computing a mean is doing statistics.

Statistical modeling

Subset that creates mathematical models to represent and analyze relationships within data.

It involves selecting, fitting and validating models to predict or understand patterns.

Claiming how the data were generated is building a model.

$$Y_i \sim N(\mu_i, \sigma^2) \quad \text{with} \quad \mu_i = \beta_0 + \beta_1 x$$

Every model has two parts

Random component

The distribution of the response

$$Y \sim N(\mu, \sigma^2)$$

Systematic component

How the mean depends on the explanatory variables

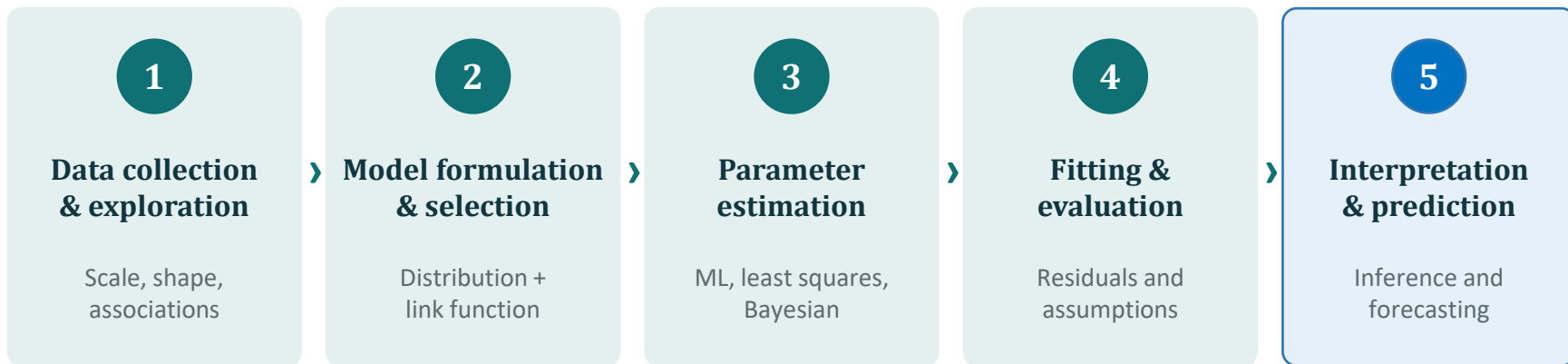
$$\mu_i = \beta_0 + \beta_1 x_i$$

The whole course generalizes these two parts

The random component becomes any distribution from the exponential family, and the systematic one connects to the mean through a link function:

$$g(\mu_i) = \mathbf{x}_i^T \boldsymbol{\beta}$$

The modeling cycle



🔄 If step 4 fails, you go back to step 2 — not to “fixing” the data.

Modeling is iterative. A model that fails validation is information, not a failure.

Always explore the data

1 Where do they come from?

Data wrangling: verify the presence of information from multiple origins, ensuring data reliability.

2 What is the scale?

Continuous or categorical? If categorical: how many categories, nominal or ordinal?

3 What is the shape?

Frequency tables, dot plots, histograms and other graphical methods.

4 How does it associate?

Cross tabulations, scatter plots, statistics by factor.

A model always produces numbers. Whether they mean anything depends on this step.

The response scale decides the method

Choosing an appropriate statistical model that fits the data.

Business question	Response	Distribution	Model
How much will we bill?	Currency (continuous)	Normal	Normal linear
Will this customer churn?	Yes / No	Bernoulli	Logistic
Which plan will they pick?	3 unordered categories	Multinomial	Nominal logistic
How will they rate us (1–5)?	Ordered levels	Multinomial	Ordinal logistic
How many tickets tomorrow?	Counts 0,1,2,...	Poisson	Poisson regression
How long will they stay?	Censored time	Exponential, Weibull	Survival

This map is what makes Module 1 valuable: the response decides the method, not the other way around.

The model does not come from the data

Knowledge that feeds the choice

- Context in which data were obtained
- Question of interest
- Theoretical relationship among variables
- Exploratory data analysis results

And it produces exactly two objects

1. A probability distribution for Y

$$Y \sim N(\mu, \sigma^2)$$

2. An equation linking $E[Y]$ with a linear combination of the explanatory variables

$$g(\mu_i) = \beta_0 + \beta_1 x_1 + \dots + \beta_p x_p$$

“Linear” means linear in the parameters β , not in the variables: $\mu = \beta_0 + \beta_1 x + \beta_2 x^2$ still qualifies.

That linearity is what turns estimation into a convex optimization problem.

Finding the parameters



Maximum likelihood

Picks the parameter that makes the observed data most probable.



Least squares

Picks β minimizing the sum of squared residuals $\sum (y_i - \hat{\mu}_i)^2$.



Bayesian analysis

Alternative approach: combines likelihood with prior information. Unit 4.

Does the model actually fit?

Regression models

- 1 MSE**
Mean squared error — penalizes big misses
- 2 RMSE**
Square root of MSE — same units as Y
- 3 MAE**
Mean absolute error — robust to outliers
- 4 R^2**
Share of variance explained by the model

Classification models

- 1 Accuracy**
 $(TP + TN) / \text{all predictions}$
- 2 Precision**
 $TP / (TP + FP)$ — of those flagged, how many were real
- 3 Recall**
 $TP / (TP + FN)$ — of the real ones, how many we caught
- 4 F1 and AUC**
Balance of the two, and ranking quality

Model assumptions — what you verify is not the fit

Linearity · independence · homoscedasticity · normality. Standardized residuals should behave like a standard normal:

$\pm 1.96 \rightarrow 5\% \text{ outside}$

$\pm 2.58 \rightarrow 1\% \text{ outside}$

Statistical metric $\neq$ business metric

Imagine you have 100,000 clients, with a 6 % churn rate, a model that predicts “nobody cancels” reaches 94 % accuracy and zero value.

False negative

Was going to churn, model said no

\$18,000

lifetime value
lost

False positive

Was not going to churn, model said yes

\$900

unnecessary
discount

The asymmetry is 20 to 1

So the classification threshold should not be 0.5. The optimal cut-off follows from the costs, not from a convention. A model with lower overall accuracy can be far more profitable.

Two objectives that are not the same

	Inference	Prediction
Question	What moves the needle?	What is going to happen?
Example	How much does house size influence price?	What is the price of this new house?
Who decides	Leadership, strategy	Operations, supply chain
What matters	Standard error of β , confidence intervals	Out-of-sample error
Winning model	The interpretable one	The one that errs least

A bank may hold a model more accurate than logistic regression and still be unable to use it: regulation requires explaining to the applicant why credit was denied.

THE MOST EXPENSIVE MISTAKE

When association does not license the conclusion

The Australian longitudinal study

Women in country areas

Fewer family physician consultations

Women near health services

More family physician consultations

Why?

Explanation A

Women in country areas are healthier and need less care.

Explanation B

Structural factors: shortage of doctors, higher visit costs, longer distances to travel.

The data are equally compatible with both. The decisions they imply are opposite.

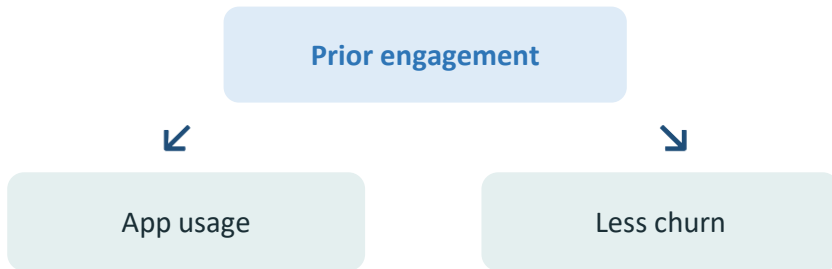
The same trap, inside your company

“Customers who use our app churn 40 % less. Let’s invest 8 million in driving downloads.”

Explanation A — the app drives retention



Explanation B — a hidden variable



The coefficient the software returns is identical in both worlds.

It is solved by design, not by more data: an A/B test randomizes who receives the promotion and breaks the arrow from prior engagement.

CASE 2 · INSURANCE PRICING

**Where GLMs are not
academic — they are the
industry standard**

Auto and property insurers price policies with generalized linear models.

When a sound model is still wrong

The model validates cleanly. Residuals behave. Out-of-sample error is excellent. And it may still be unusable.

The proxy problem

Postal code predicts claims well and is legal to use. It also tracks ethnicity, because of how cities were built. Price on it and you charge one group more — without ever touching the forbidden variable.

Predictive power is not a defense.

Interpretability is how you catch it

Detect — you see how much weight it carries

Test — you measure what removing it costs

Defend — you can explain the rate to a regulator

In a black box the proxy is still there — invisible.

A different failure mode from Case 1

Case 1 was a mistake — you read association as causation. This is not a mistake at all: the statistics are correct and the conclusion is still unacceptable. Step 5 is where you judge whether a coefficient should be acted on, not just whether it is significant.